

1 Agroforestry SmartAgroforst: Poplar biomass modelling

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3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

6 1. Introduction

7 Agroforestry systems that integrate trees with agricultural crops are increasingly
8 recognised as a promising land-use option to reconcile climate mitigation, biodiversity
9 conservation, and rural income generation on the same hectare of land [10, 14]. Temper-
10 ate short-rotation agroforestry systems (SRAFS) based on fast-growing species such as

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11 poplar, willow, alder, and black locust can deliver substantial biomass yields while im-
12 proving soil quality and providing multiple ecosystem services compared to conventional
13 monocultures [7, 15].

14 In the last two decades, a large body of research has analysed biomass supply chains
15 for bioenergy and biorefineries using mathematical programming models, often in the
16 form of mixed-integer linear programming (MILP) [16, 12]. These models typically opti-
17 mise network configuration, feedstock sourcing, and logistics under cost or profit objec-
18 tives. They often treat biomass availability as an exogenous input or rely on simplified
19 yield functions that neglect stand growth dynamics and age-related constraints. More-
20 over, downstream processing of biomasses is fairly simplified by assuming a set of con-
21 sumption sites reflecting e.g. bioenergy plants consuming all produced biomass quantities.
22 Recent work has started to integrate operational details such as various biomass types,
23 biomass growth models and regeneration processes in multi-period supply chain models.
24 Yet most applications focus on grassland or forest residues rather than agroforestry sys-
25 tems with repeated coppice cycles and strong interactions between stand structure and
26 harvesting decisions [3].

27 Short-rotation agroforestry systems exhibit pronounced temporal variability in biomass
28 production due to species-specific growth trajectories, competition, and management
29 practices, which has been documented in long-term trials in Central Europe [7, 15]. For
30 poplar, willow, alder, and black locust, allometric tree biomass models based on stem
31 base diameter (SBD) or similar biometric variables provide a compact and empirically
32 grounded link between stand structure and aboveground woody biomass at the tree and
33 stand level [6]. These models allow the derivation of age-dependent yield coefficients for
34 different product types, but they are rarely coupled with tactical and strategic supply
35 chain optimisation. From a cleaner production perspective, it is particularly relevant to
36 capture the cascading use of biomass along quality gradients (e.g., chemical feedstocks,
37 pulp, and energy), as this can significantly affect both economic returns and environmen-
38 tal outcomes [2, 8].

39 The present paper contributes to this literature in four ways. First, we propose a
40 novel MILP formulation for agroforestry biomass supply chain design that explicitly links
41 consecutive harvest cycles through age-lag constraints and binary path variables, ensuring

that harvesting decisions at each site are consistent with minimum and maximum rotation
ages over multiple cycles. Second, we embed allometric yield functions for different
biomass types into the model by translating age-specific biomass increments into site- and
product-specific yield parameters, thereby capturing species- and management-dependent
growth dynamics in an optimisation-compatible way [6, 15]. Third, we model a product
cascading hierarchy in which higher-quality products can satisfy lower-quality demands,
which is relevant for cleaner production strategies that prioritise material uses before
energy recovery [2, 8]. Finally, we outline a computational experimentation framework
for a regional case study in Central Europe that will test the sensitivity of optimal system
design to rotation length, allometric parameter uncertainty, price and policy scenarios,
and infrastructure options.

The remainder of the paper is structured as follows. Section @ref(sec:litreview) re-
views the relevant literature on agroforestry biomass systems, allometric biomass mod-
elling, and biomass supply chain optimisation. Section @ref(sec:allometrics) introduces
the allometric models adopted to represent biomass growth of different tree species and
biomass types. Section @ref(sec:milp) summarises the MILP formulation for the agro-
forestry supply chain with age-lag constraints. Section @ref(sec:experiments) outlines the
planned computational experiments for the case study and discusses expected insights
for cleaner production strategies.

2. Literature review

Agroforestry biomass systems, tree and stand biomass models, and biomass sup-
ply chain optimisation have all been studied extensively, but they are rarely combined
in an integrated modelling framework that links age-dependent stand dynamics to re-
gional product-cascading value chains [16, 12, 5]. This section reviews (i) agroforestry
and short-rotation systems with emphasis on yields and economics, (ii) allometric and
age-dependent biomass yield models, and (iii) biomass supply chain optimisation and
cascading use of biomass, and concludes by identifying the modelling gap addressed in
this paper.

70 2.1. Biomass supply chain optimisation

71 Biomass supply chain design and planning have been widely studied using MILP
 72 models at strategic, tactical, and operational levels Zahraee et al. [16]; Sharma et al.
 73 [12]; Espinoza et al. [5]. Comprehensive reviews classify models according to decision
 74 scope (location, capacity, routing, inventory), time horizon, uncertainty treatment, and
 75 sustainability metrics, and show a strong emphasis on bioenergy and biorefinery systems
 76 Lamers et al. [8]; Espinoza et al. [5]. Typical models optimise the number, size, and
 77 locations of processing facilities, biomass sourcing patterns, multimodal transport flows,
 78 and inventory policies under cost- or profit-oriented objectives, sometimes complemented
 79 by greenhouse gas emissions or other environmental indicators Eksioglu et al. [4]; Arabi
 80 et al. [1].

81 A key methodological feature of many biomass supply chain models is the integration
 82 of long-term location and capacity decisions with multi-period logistics decisions such as
 83 harvesting, transport, storage, and processing Eksioglu et al. [4]; Zahraee et al. [16]. For
 84 example, mixed-integer formulations have been used to simultaneously determine biore-
 85 finery locations, their capacities, biomass collection and storage sites, and time-dependent
 86 flows of biomass and biofuel, explicitly accounting for seasonality, deterioration, and in-
 87 ventory trade-offs Eksioglu et al. [4]. These models demonstrate that a significant share
 88 of biofuel production costs is attributable to logistics, and that integrated optimisation
 89 of network design and operations can substantially reduce costs compared to decoupled
 90 decision making Eksioglu et al. [4]; Arabi et al. [1].

91 However, in virtually all of these models, biomass availability over time is treated
 92 as an exogenous input, either as annual supply potentials per region or as simple time
 93 profiles based on harvest windows and assumed yields Zahraee et al. [16]; De Meyer
 94 et al. [3]. Stand-level growth dynamics, age-class structure, and rotation decisions are
 95 not represented explicitly; instead, land-use or yield scenarios enter as fixed parameters
 96 De Meyer et al. [3]; Lamers et al. [8]. As a result, these models cannot explore trade-offs
 97 between rotation length, stand management, and supply chain configuration in a unified
 98 optimisation framework, especially not for agroforestry systems with repeated coppice
 99 cycles and strong interactions between tree growth and harvesting decisions.

100 *2.2. Agroforestry and short-rotation systems*

101 Agroforestry deliberately integrates woody perennials with crops and/or livestock on
102 the same management unit, with the aim of enhancing productivity, profitability, and en-
103 vironmental performance Nair [10]; Toensmeier [14]. In temperate regions, short-rotation
104 agroforestry systems (SRAFS) often take the form of alley-cropping, where double or
105 multiple tree rows alternate with crop strips, enabling simultaneous production of agri-
106 cultural and woody biomass Huber et al. [7]. Empirical trials in Southern Germany and
107 Central Europe show that SRAFS with poplar, willow, black alder, and black locust can
108 achieve mean annual increments (MAI) of roughly 7–10 t per ha and year in aboveground
109 woody biomass during the first rotation, with pronounced species-specific differences in
110 growth trajectories, stand structure, and mortality (see Huber et al. [7]; Trnka et al.
111 [15]).

112 For short-rotation coppice (SRC) and very short-rotation coppice (vSRC) poplar plan-
113 tations in Italy and Central Europe, economic analyses indicate that rotations of about
114 5–7 years at moderate planting densities can be competitive with arable crops, but prof-
115 itability is highly sensitive to yield assumptions, establishment costs, and logistics [13].
116 This sensitivity implies to couple stand-level growth and models with supply chain optimi-
117 sation models, rather than treating biomass availability as an exogenous, time-invariant
118 parameter [16]. Beyond yields and economics, agroforestry and SRC systems have been
119 widely analysed for carbon sequestration, nutrient cycling, and biodiversity, reinforcing
120 their role as candidate feedstock systems for regional bioeconomy strategies that empha-
121 sise material-before-energy use of biomass (see e.g. Toensmeier [14]).

122 Thus, a fundamental cornerstone of AFS modeling is the representation of biomass
123 growth dynamics, which can be achieved through allometric models that link tree bio-
124 metric variables to biomass, and through age-dependent stand biomass functions that
125 capture growth trajectories over rotation cycles. These models provide the necessary
126 link between stand structure and biomass availability, which can then be integrated into
127 supply chain optimisation models to explore trade-offs between rotation length, product
128 portfolios, and infrastructure configuration in AFS supply chains.

129 2.3. Tree-level allometric models

130 Allometric tree biomass models relate easily measured variables, such as stem base di-
 131 ameter (SBD) or diameter at breast height, to aboveground dry biomass using power-law
 132 functions of the form $M = b_0 \text{SBD}^{b_1}$, with species-specific coefficients b_0 and exponents b_1
 133 [6]. These models allow non-destructive estimation of individual-tree and stand biomass
 134 from diameter measurements and have been widely used in forestry and agroforestry for
 135 both carbon accounting and management planning [6, 14].

136 For SRAFS in Southern Germany, species-specific allometric functions based on SBD
 137 measured 10 cm above ground were developed for black alder, black locust, poplar (Max
 138 3), and willow (Inger) Huber et al. [6]. The reported models share a common exponent
 139 $b_1 = 2.603$ across species, while the scale parameter b_0 ranges from about 0.025 to 0.041,
 140 reflecting differences in wood density and branching pattern Huber et al. [6]. Applying
 141 these functions to all shoots per tree and aggregating over observed tree densities yields
 142 annual aboveground leafless dry biomass at plot and stand level, from which current
 143 annual increment (CAI) and MAI curves can be derived for each species and management
 144 system Huber et al. [7].

145 These SRAFS studies demonstrate that stand structure, diameter distributions, and
 146 mortality strongly affect aggregate yields, with black locust showing high wood density
 147 and strong size inequality and willow showing high shoot densities but low wood density
 148 and diameter growth Huber et al. [7]. From a modelling perspective, the key point is
 149 that tree-level allometrics provide a robust mapping from biometric state variables to
 150 biomass, which can be exploited either to build detailed stand simulators or to construct
 151 age-class yield tables for optimisation models Huber et al. [6].

152 2.4. Age-dependent biomass yield functions

153 Existing allometric models link tree biomass to diameter or height but do not yet
 154 provide **stand-level** biomass as an explicit function of stand age. Age-dependent biomass
 155 functions are useful for yield projections in short-rotation forestry and agroforestry, where
 156 management is organized in discrete rotations (e.g. 5–15 years). [11, 13]

157 A common approach is to derive age functions from rotation trials that report mean
 158 annual increment (MAI) over time and identify the age at which MAI plateaus. For

159 poplar SRWC in northern Poland, Niemczyk et al. (2021) showed that MAI in above-
 160 ground dry biomass increases up to about age 10; beyond this, the MAI curve flattens,
 161 suggesting a biologically optimal rotation age around 10 years because the slope of MAI
 162 is near zero. [11] Reported MAI values for single-stem poplar plantations in a 10-year
 163 rotation ranged approximately from 1–15 Mg DM ha⁻¹ yr⁻¹ depending on cultivar. [11]
 164 Similar magnitudes and rotation lengths (about 5–7 years for SRC, up to 10–15 years for
 165 SRC/SRF) are reported for poplar SRC systems in Italy, where whole-rotation averages
 166 of roughly 15 Mg DM ha⁻¹ yr⁻¹ are considered economically viable. [13]

167 A simple and widely used parametric form for age-dependent stand biomass $B(t)$ is
 168 the Chapman–Richards (or Mitscherlich/monomolecular) function:

$$B(t) = B_{\max} (1 - e^{-k \cdot t})^m$$

169 with stand age t in years, asymptotic biomass $B_{\max}$ (Mg DM ha⁻¹), rate parameter k
 170 (year⁻¹), and shape parameter m . This formulation captures the rapid juvenile growth,
 171 a phase of approximately linear biomass accumulation, and a saturating phase where
 172 growth slows. [11] The corresponding current annual increment (CAI) and mean annual
 173 increment (MAI) are

$$\text{CAI}(t) = \frac{dB}{dt}, \quad \text{MAI}(t) = \frac{B(t)}{t}.$$

174 The age maximizing MAI (optimal rotation under a volume/biomass criterion) can be
 175 derived analytically or determined numerically. Fitting such functions to observed age-
 176 biomass or age-volume series from rotation trials allows cultivar-specific yield curves that
 177 are directly usable in agroforestry simulation models. [11, 13]

178 For illustration, we parameterize three stylized poplar curves representing low, medium,
 179 and high productivity on temperate mineral soils. Parameter values are loosely informed
 180 by the ranges reported by Niemczyk et al. (2021) and Testa et al. (2014) and should be
 181 recalibrated when site-specific data are available. [11, 13]

- 182 • Low productivity: $B_{\max} = 80$ Mg DM ha⁻¹, $k = 0.18$, $m = 1.1$
- 183 • Medium productivity: $B_{\max} = 120$ Mg DM ha⁻¹, $k = 0.20$, $m = 1.2$
- 184 • High productivity: $B_{\max} = 160$ Mg DM ha⁻¹, $k = 0.22$, $m = 1.3$

185 These values yield MAI maxima between about 8 and 12 years with peak MAI values
 186 of roughly 8–16 Mg DM ha⁻¹ yr⁻¹, consistent with the empirical range where 10-year

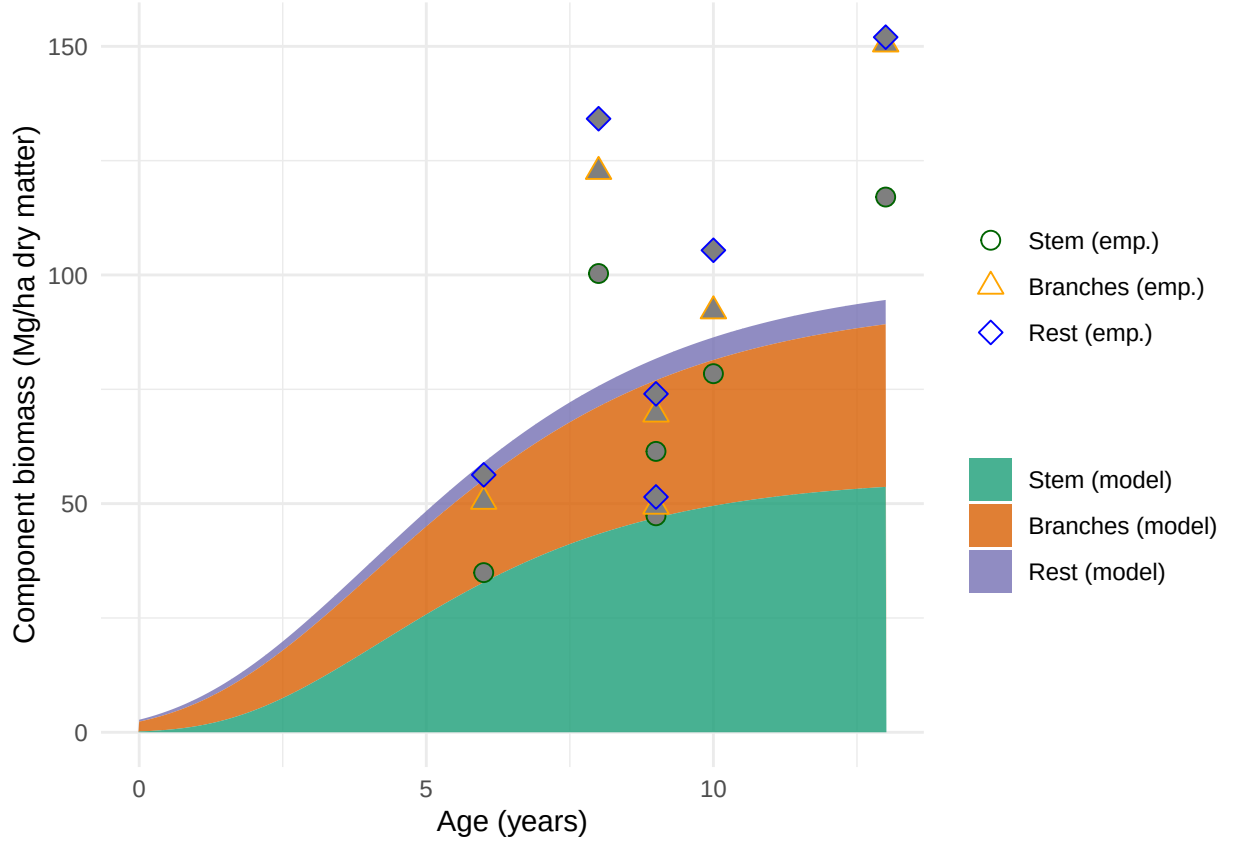


Figure 1: Modeled stand-level biomass components for poplar clone Max 3 over age, represented as stacked Gompertz curves and compared with empirical reference values from agroforestry trials.

187 rotations produced MAI of about 10–15 Mg DM ha⁻¹ yr⁻¹ for the best poplar cultivars
 188 in northern Poland and Italy. [11, 13]

189 In the agroforestry context, such age functions can be combined with tree density and
 190 allometric partitioning to above- and below-ground biomass to obtain tree-row biomass
 191 over time for different designs. Calibration against local inventory data or growth mea-
 192 surements (e.g. basal area increment and height growth) is recommended before using
 193 the curves for scenario analysis.

194 2.5. Cascading use of biomass and product quality

195 From a cleaner production perspective, cascading use of biomass—prioritising high-value
 196 material uses before energy recovery—is widely advocated to maximise value creation

and reduce environmental burdens Lamers et al. [8]; Espinoza et al. [5]. In wood-based value chains, this typically implies allocating higher-quality stem fractions to products such as sawn timber, veneer, panels, or pulp, while directing lower-quality fractions and residues to energy applications De Meyer et al. [2]. Multi-product MILP models such as OPTIMASS implement this concept by defining a hierarchy of product quality classes and allowing higher-quality products to satisfy lower-quality demands via substitution constraints along the cascade De Meyer et al. [2]; De Meyer et al. [3].

For agroforestry biomass such as poplar and willow, this logic translates into a set of product streams (e.g. larger-diameter logs for material uses, medium-diameter fractions for pulp and panels, small-diameter wood and residues for energy), whose relative shares depend on stand age and growth conditions Huber et al. [7]; Trnka et al. [15]. In the present work, the age-class yield construction described above is extended by associating each age class with product-specific yield fractions, thereby capturing age-dependent quality in a simplified but optimisation-compatible form. The MILP formulation then enforces a product hierarchy in which higher-quality products can satisfy lower-quality demands, subject to the available yields and infrastructure capacities, thus embedding a simplified cascading logic into the regional supply chain design De Meyer et al. [2].

2.6. *Synthesis and modelling gap*

The literature summarised above provides three critical ingredients for modelling agroforestry biomass supply chains. First, empirical SRAFS and SRC studies offer species- and site-specific information on growth, MAI, stand structure, and economic performance, including allometric models that link diameter distributions to biomass Huber et al. [6]; Huber et al. [7]; Trnka et al. [15]; Testa et al. [13]. Second, age-dependent stand biomass functions such as Chapman–Richards provide a convenient parametric representation of growth over rotation, from which discrete age-class yields can be derived Niemczyk et al. [11]. Third, biomass supply chain MILP models and multi-product cascading formulations offer a rich toolkit for optimising infrastructure, logistics, and product allocation in regional bioeconomy systems Zahraee et al. [16]; De Meyer et al. [2]; De Meyer et al. [3]; Espinoza et al. [5].

At the same time, existing models fall short of explicitly coupling agroforestry stand dynamics with supply chain design in several respects. Most biomass supply chain for-

228 mulations assume exogenous biomass availability profiles and do not represent stand age
 229 or rotation decisions, while forestry-oriented growth and yield models are rarely embed-
 230 ded in regional optimisation models that include multiple products and cascading uses
 231 De Meyer et al. [3]; Lamers et al. [8]. In particular, there is a lack of models that (i)
 232 link age-dependent stand biomass functions to age-class yield tables for different product
 233 types, (ii) enforce multi-cycle age-lag constraints that ensure consistency of harvesting
 234 decisions at each agroforestry site across successive rotations, and (iii) allocate biomass
 235 along a product quality cascade within a spatially explicit, multi-period supply chain
 236 optimisation framework De Meyer et al. [2]; De Meyer et al. [3]; Eksioglu et al. [4].

237 The MILP formulation developed in this paper addresses this gap by integrating
 238 age-dependent biomass yield modelling, stand-level age-lag constraints, and a prod-
 239 uct-cascading hierarchy into a single optimisation model for the design and operation
 240 of agroforestry biomass supply chains.

241 3. Biomass supply chain modelling

242 Biomass supply chain design and planning has been widely studied using MILP mod-
 243 els at strategic, tactical, and operational levels [16, 12]. Comprehensive reviews classify
 244 models according to decision scope (location, capacity, routing, inventory), time hori-
 245 zon, uncertainty treatment, and sustainability metrics [16, 5]. Many models focus on
 246 bioenergy supply chains for electricity, heat, or biofuels, considering feedstocks such as
 247 agricultural residues, forestry residues, dedicated energy crops, and municipal wastes [12].
 248 Environmental aspects, including greenhouse gas emissions and land use, are increasingly
 249 integrated, often using life-cycle assessment (LCA) data or emission factors [5].

250 Among generic models, OPTIMASS is a notable example of a multi-product MILP
 251 that optimises biomass supply chains with changing biomass characteristics and by-
 252 product re-injection [2]. It models the flow of multiple biomass types and products
 253 through supply chain stages, capturing conversion yields, storage, and cascading uses.
 254 The t-OPTIMASS extension further incorporates biomass growth and regeneration over
 255 multiple periods, enabling the representation of time-dependent availability for biomass
 256 supply chains [3]. However, these formulations generally assume exogenous biomass
 257 availability profiles per period, and do not explicitly model stand-level age dynamics or

258 rotation constraints.

259 Several scenario-based approaches have been developed for forest biorefinery supply
260 chains, linking product portfolios, process choices, and supply chain design under mar-
261 ket volatility [9]. Lamers et al. [8] discuss strategic feedstock supply system design for
262 biorefineries, emphasising the importance of economies of scale, conversion yields, and
263 logistics in achieving cost-efficient and robust systems. Reviews highlight that only a
264 limited subset of models explicitly consider biomass growth, spatially explicit land-use
265 change, or agroforestry systems [16, 12]. There is therefore a clear opportunity to de-
266 velop models that couple agroforestry stand dynamics with network design and cleaner
267 production objectives.

268 3.1. Cascading use of biomass

269 From a cleaner production perspective, the cascading use of biomass—prioritising
270 high-value material uses before energy recovery—is widely advocated to maximise re-
271 source efficiency and mitigate environmental impacts [8]. In wood-based value chains,
272 this implies that high-quality timber or chemical feedstock fractions should first be al-
273 located to material products such as engineered wood products, pulp and paper, or
274 biochemicals, while lower-quality fractions and residues are used for energy [2, 8]. MILP
275 models can represent such cascades by defining product quality classes and allowing
276 substitution of higher-quality products for lower-quality demands, subject to hierarchy
277 constraints.

278 The OPTIMASS framework explicitly models such product cascades by defining a
279 product quality order and allowing flows from higher-quality products to lower-quality
280 demand nodes [2]. Similar ideas have been applied in forest-based biorefinery supply
281 chains, where product portfolios include platform chemicals, fuels, and electricity [9]. For
282 agroforestry biomass, poplar and willow can supply different product streams, including
283 sawlogs, peeled veneers, pulpwood, and energy chips, with quality primarily defined by
284 diameter, straightness, and defect rates [13?]. Economic analyses in Italy and Central
285 Europe suggest that the economic viability of SRC systems depends on access to higher-
286 value markets such as panel products and pulp, in addition to energy markets [13].

287 Cleaner production strategies that integrate agroforestry biomass into regional bioe-
288 conomies therefore require models that can allocate biomass flows along such cascades

while respecting stand growth dynamics, rotation constraints, and logistical capacities. Explicit representation of age-dependent yield quality (e.g., diameter distributions, wood density) remains challenging in large-scale MILP models, but can be approximated through age- and species-specific product yield fractions.

3.2. Stand-level yield functions and age classes

To integrate allometric growth into a MILP formulation, we abstract from individual trees and construct age-class yield functions for each biomass type p and age a . Following common practice in forest and SRC modelling, we define discrete annual age classes and compute the stand biomass yield Y_{pa} (t ha^{-1}) associated with harvesting at age a by combining allometric functions with empirically observed diameter distributions and stand densities [7, 15]. Where detailed stand structure data are available, this can be done by simulating SBD distributions over time (e.g., via mixed-effects models) and applying tree-level allometrics to each diameter class [7]. In the absence of such detailed data, age-dependent MAI and CAI curves reported in the literature can be used to approximate Y_{pa} [15, 7].

In the context of the agroforestry supply chain design (SCD) model, we assume that each agroforestry site i is characterised by a species or species mix, a planting density, and management regime, which jointly determine a site-specific yield profile η_{pa} (t ha^{-1}) for each product type p at age a . These yield profiles can be interpreted as the maximum harvestable biomass per hectare for product p if the stand is harvested at age a , conditional on the stand having been established at an earlier time. In the current MILP formulation, these profiles are captured through the parameter η_{pa} , termed base yield, and are linked to decision variables via the constraint

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i, p, t \in \mathcal{T}^{\text{harv}}.$$

Here, Y_{ipt} is the maximum harvest quantity of product p at site i in period t , AREA_i is the site area (ha), and z_{ist} is a binary variable indicating that harvests at site i in periods s and t form consecutive harvests with an age lag $t - s$ within allowed bounds. The term $\eta_{p(t-s)}$ thus plays the role of an allometrically derived yield coefficient that depends only on stand age $a = t - s$, not on calendar time.

317 3.3. Biomass types and product quality classes

318 In the model, three product types are distinguished: $p = 1$ (chemical-grade biomass),
 319 $p = 2$ (pulp-grade biomass), and $p = 3$ (energy-grade biomass). This tripartite classi-
 320 fication reflects a product quality hierarchy, with product 1 having the highest quality
 321 and price, and product 3 being the lowest-quality fraction typically used for energy [2, 8].
 322 From an allometric perspective, the differentiation among these products can be linked
 323 to diameter classes, stem straightness, and defect rates, as these determine the suitability
 324 of biomass fractions for different processing routes [13?].

325 In practice, allometric models provide estimates of total aboveground woody biomass,
 326 which can be split into product classes using empirical conversion factors or allocation
 327 rules derived from product recovery studies (e.g., share of biomass suitable for veneer
 328 vs. pulp vs. chips as a function of diameter and log length) [?]. For the purposes of
 329 the SCD model, we can represent this by specifying product-specific yield coefficients η_{pa}
 330 that sum to the total biomass yield at age a and reflect the expected proportion of each
 331 product class under a given management regime.

332 The cascading hierarchy in the MILP formulation is implemented through demand
 333 satisfaction constraints that allow higher-quality products to satisfy lower-quality de-
 334 mands. Specifically, demand $D_{kpt}^{\max}$ at consumer k for product class p in period t can
 335 be met by flows of product p and all higher-quality classes $p' \leq p$ from storage hubs, as
 336 captured by

$$\sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k, p, t.$$

337 This structure supports cleaner production by favouring material uses when possible and
 338 relegating residual fractions to energy uses, depending on revenue parameters $R_{k,p}$ and
 339 system constraints [2, 8].

340 4. MILP model for agroforestry supply chain design

341 4.1. Sets, indices, and time structure

342 The model considers a set of agroforestry sites $\mathcal{I}$, storage/processing facilities $\mathcal{J}$,
 343 consumer sites $\mathcal{K}$, and product types $\mathcal{P} = \{1, 2, 3\}$. Time is discretised into annual periods
 344 $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To represent stand establishment and harvest timing, extended

time sets $\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$ and harvest periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$ are defined, where $A^{\min}$ and $A^{\max}$ are minimum and maximum stand ages (years) at harvest. The set $\mathcal{S}$ comprises all ordered pairs (s, t) of consecutive harvest periods (including establishment at $s = 0$) satisfying the age-lag bounds:

$$\mathcal{S} = \{(s, t) \mid s, t \in \{0, \dots, T_{\max} + 1\}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}.$$

4.2. Decision variables

Binary variables z_{ist} indicate whether, at site i , periods s and t form consecutive harvests (or establishment and first harvest if $s = 0$), with $(s, t) \in \mathcal{S}$. These variables define a path of consecutive harvest cycles at each site over the planning horizon. Continuous variables $Y_{ipt} \geq 0$ represent the maximum harvest quantity (t) of product p at site i in harvest period t . Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from sites to storage and from storage to consumers, respectively, and S_{jpt} represent inventories at storage locations.

4.3. Objective function

The model maximises net present value over the planning horizon, expressed as total revenues minus establishment, opportunity, harvest, transport, and storage costs:

$$\begin{aligned} \max Z = & \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkpp't} \\ & - \sum_i C_i^{\text{est}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} z_{i0t} \\ & - \sum_i C_i^{\text{opp}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} (T_{\max} - t) \cdot z_{i0t} \\ & - \sum_{i, (s,t) \in \mathcal{S}: s > 0} C_i^{\text{harv}} \cdot \text{AREA}_i \cdot z_{ist} \\ & - c^{\text{tr-raw}} \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} - c^{\text{tr-pre}} \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} - \sum_{j,p,t} c_j^{\text{stor}} \cdot S_{jpt}. \end{aligned}$$

Here, $R_{k,p}$ are product- and consumer-specific revenues (€/t), C_i^{est} are establishment costs per hectare, C_i^{opp} are annual opportunity costs per hectare, and C_i^{harv} are harvest and refitting costs per hectare [13]. Transport costs are proportional to flow and distance with rates $c^{\text{tr-raw}}$ and $c^{\text{tr-pre}}$ for raw and pre-treated biomass, respectively, and storage incurs linear holding costs c_j^{stor} per tonne.

365 4.4. Key constraints

366 The path connectivity constraint ensures that each harvest period at a site has exactly
 367 one predecessor and one successor in the age-lag graph, effectively forming a path over
 368 $\mathcal{T}^+$:

$$\sum_{s=0:(s,t) \in \mathcal{S}}^t z_{ist} = \sum_{u=t+1:(t,u) \in \mathcal{S}}^{T_{\max}+1} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}.$$

369 Establishment is restricted to at most one time per site:

$$\sum_{t \in \mathcal{T}^+} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I}.$$

370 Biomass yield is linked to age-lag decisions through the allometric yield coefficients:

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\text{harv}}.$$

371 Harvest quantities bound outbound flows from each site:

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i, p, t \in \mathcal{T}^{\text{harv}}.$$

372 At storage hubs, inventories obey standard balance equations with initial inventory
 373 zero:

$$S_{jpt} = S_{jp,t-1} + \sum_i X_{ijpt} - \sum_{k,p' \geq p} X_{jkpp't} \quad \forall j, p, t \in \mathcal{T}^{\text{harv}}.$$

374 Storage and processing capacities are enforced via:

$$\sum_p S_{jpt} \leq \text{CAP}_j^{\text{stor}} \quad \forall j, t, \quad \sum_{i,p} X_{ijpt} \leq \text{CAP}_j^{\text{proc}} \quad \forall j, t.$$

375 Demand satisfaction with cascades is modelled as described above, where higher-
 376 quality products can satisfy lower-quality demands up to maximum demand levels $D_{kpt}^{\max}$.

377 5. Case study and computational experiment plan

378 The proposed MILP model is intended to support the design of agroforestry biomass
 379 supply chains for cleaner production in specific regional contexts. In a forthcoming case
 380 study, we focus on poplar-based alley-cropping systems in a temperate European setting,
 381 informed by empirical data from SRAFS trials in Southern Germany and SRC plantations
 382 in Central Europe [7, 15, 13].

383 We envisage the following steps and experimental scenarios for computational analy-
384 sis:

- 385 1. **Parameterisation of allometric yield profiles:** Use species- and site-specific
386 allometric models and observed growth trajectories to construct age-dependent
387 yield coefficients η_{pa} for chemical, pulp, and energy product classes at representative
388 agroforestry sites. Where necessary, scenario ranges will be defined to capture
389 uncertainty in MAI and CAI due to climate variability and management differences
390 [6, 7, 15].
- 391 2. **Baseline optimisation:** Solve the MILP model for a baseline configuration with
392 fixed prices, capacities, and policy settings, obtaining optimal establishment timing,
393 harvest schedules, and supply chain structure (locations, flows, inventories).
- 394 3. **Rotation length scenarios:** Vary the age-lag bounds $A^{\min}$ and $A^{\max}$ to represent
395 alternative rotation strategies (e.g., 3–5 years vs. 8–10 years) and analyse their
396 impacts on profitability, product mix, and land use. This will shed light on trade-
397 offs between shorter rotations favouring energy chips and longer rotations enabling
398 higher shares of material products [15, 13].
- 399 4. **Product price and cascading scenarios:** Explore scenarios with different rel-
400 ative prices for chemical, pulp, and energy products, reflecting changes in market
401 demand and policy incentives (e.g., bioenergy subsidies, green chemicals markets).
402 Assess how the product cascade hierarchy and yield quality structure influence the
403 marginal value of species and rotation decisions [2, 8].
- 404 5. **Infrastructure and logistics scenarios:** Evaluate the effect of alternative stor-
405 age and processing network configurations (e.g., decentralised vs. centralised hubs,
406 different capacity levels, transport cost parameters) on the optimal spatial deploy-
407 ment of agroforestry sites and biomass flows [2, 3, 8].
- 408 6. **Sensitivity to allometric and growth uncertainty:** Conduct parametric sen-
409 sitivity and, where computationally feasible, stochastic analyses to quantify the
410 impact of uncertainty in allometric parameters and growth responses (e.g., under
411 drought or management differences) on optimal decisions and system robustness
412 [7, 15].
- 413 7. **Cleaner production indicators:** For selected solutions, compute derived indica-

tors such as biomass cascading index (share of biomass used for material products before energy), utilisation rates of higher-quality fractions, and, where coupled with LCA data, greenhouse gas emission savings relative to fossil benchmarks [5, 16].

6. Conclusions

This paper outlines a research framework that couples allometric biomass growth models with a MILP-based agroforestry biomass supply chain design model featuring age-lag constraints and product cascading. Building on empirical evidence from temperate SRAFS and SRC systems and established biomass supply chain models, the proposed approach aims to capture key interactions between stand-level growth dynamics and network-level design decisions that are central to the cleaner production potential of agroforestry [7, 2, 3]. Future work will implement and calibrate the model for a regional case study, conduct the computational experiments sketched above, and extend the framework towards multi-objective optimisation including environmental indicators such as greenhouse gas emissions and biodiversity proxies.

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